Introduction to Circuits & Ohm's Law

Name: _____

Date: _____ **Period:** _____

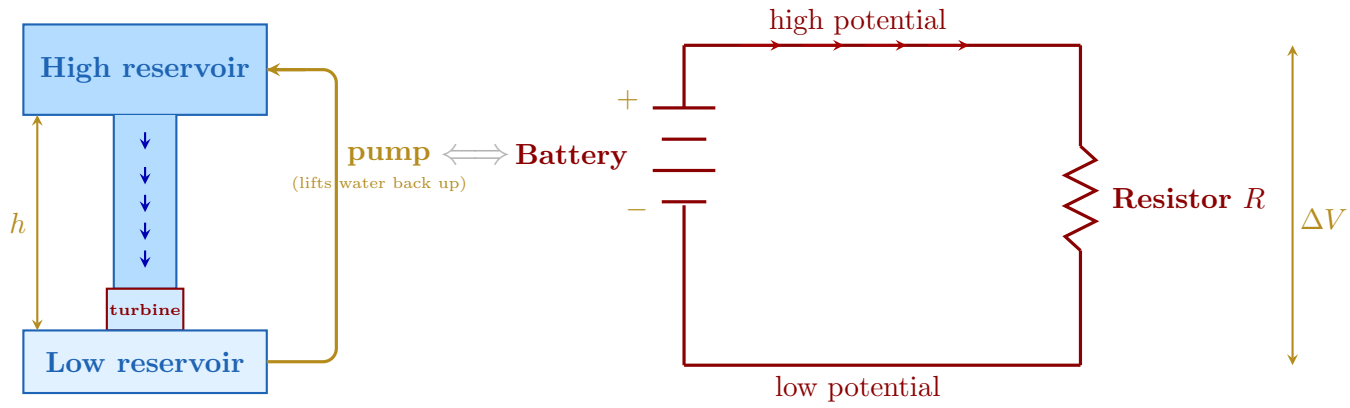
0 WARM-UP

A ball sits at the top of a hill. A creek flows down the same hill.

What does each one *have*, and what is the key difference between them?

1 THE WATERFALL ANALOGY

Follow along as your teacher draws on the board. Fill in the table below.



The Waterfall–Circuit Analogy

Water System	Electric Circuit	SI Unit
Height difference h	_____ (potential difference)	_____
Flow rate (vol./time)	_____ (charge/time)	_____
Narrow pipe	_____	_____
Pump	_____	—

Key connection to mechanics: Gravitational PE per unit mass $= gh$.

Similarly, **voltage** is _____ per unit _____.

$$\Delta V = \frac{\Delta U_E}{\text{_____}}, \qquad m\,g\,h \longleftrightarrow q\ \text{_____}$$

Prediction: If I increase the waterfall height (= voltage) while the pipe width (= resistance) stays fixed, what happens to the flow rate (= current)?

2 CURRENT, VOLTAGE, RESISTANCE, AND OHM'S LAW

Complete the definitions as we work through them together.

The Three Core Quantities

1. **Current** I: rate of _____ flow past a point.

$$I = \frac{\Delta _____}{\Delta _____} \qquad \text{Units: } _____ \quad (\text{symbol: } _____)$$

Water analogy: $I \leftrightarrow$ _____

2. **Voltage** ΔV: _____ per unit _____.

$$\Delta V = \frac{\Delta U_E}{_____} \qquad \text{Units: } _____ \quad (\text{symbol: } _____)$$

Water analogy: $\Delta V \leftrightarrow$ _____

3. **Resistance** R: opposition to _____ flow.

$$\text{Units: } _____ \quad (\text{symbol: } _____) \; = \; \frac{_____}{_____}$$

Water analogy: $R \leftrightarrow$ _____

Ohm's Law

$$\boxed{V = I \cdot R}$$

Rearrange to solve for I and for R:

$$I = _____ \qquad R = _____$$

Note: Ohm's Law is *empirical*, not a universal law of nature. Materials that obey it are called _____; those that do not are called _____.

Quick-Check Problems

1. A resistor with $R = 10\,\Omega$ has $V = 5\,\text{V}$ across it. Find I.

Answer: _____

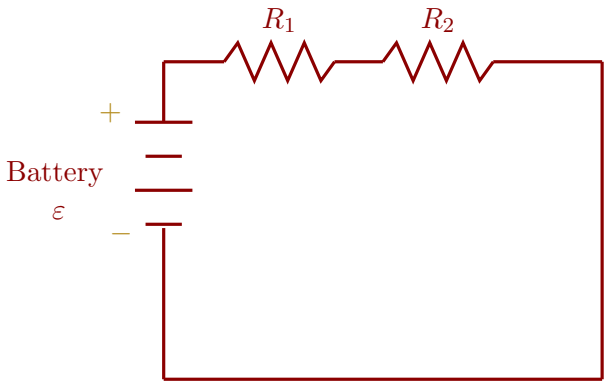
2. A current $I = 2\,\text{A}$ flows through a $6\,\Omega$ resistor. Find V.

Answer: _____

3 SERIES AND PARALLEL CIRCUITS

Two ways to connect resistors — with very different results.

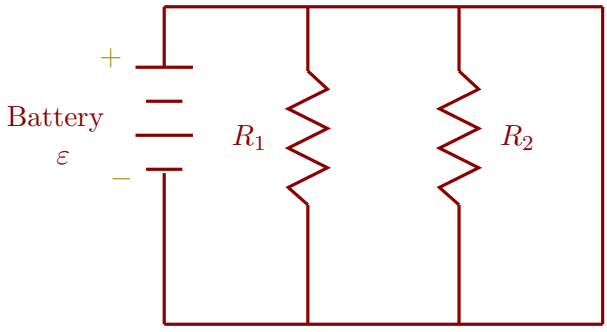
Series



$$R_{\mathbf{eq}} = R_1 + R_2$$

same _____ through each

Parallel



$$\frac{1}{R_{\mathbf{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

same _____ across each

Series vs. Parallel Summary

	Series	**Parallel**
R_{eq} formula	_____	_____
Quantity that is the *same* for each R	_____	_____
Adding more resistors	R_{eq} goes _____	R_{eq} goes _____

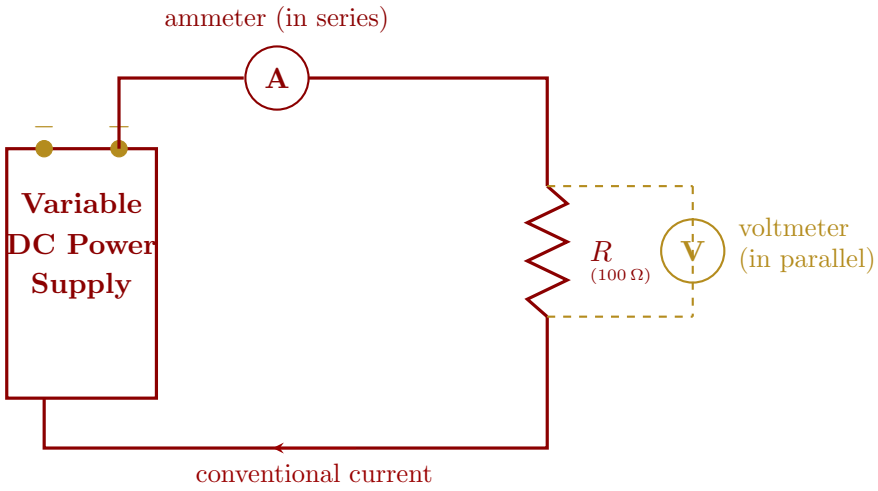
Water analogy check: Consider two configurations of two narrow pipes: side-by-side (parallel) vs. end-to-end (series). Which lets more water flow? Explain in terms of resistance.

Draw the equivalent circuit for each case (replace R_1 and R_2 with a single R_{eq}):

4 LAB: OHM'S LAW EXPLORATION

Goal: Build a simple DC circuit with a 100 Ω resistor. Increase the supply voltage from 1 V to 8 V, record the resulting current, plot V vs. I, and extract the resistance from the slope of your graph.

Circuit Diagram



Key setup rules:

- **Ammeter (A) goes in series** — it physically breaks one wire so *all* current flows through it.
- **Voltmeter (V) goes in parallel** — clip its probes to the two ends of the resistor without breaking the main wire.
- **Do NOT turn on the supply until your teacher has checked your circuit.**
- **Safety:** When probing a live circuit, keep **one hand in your pocket**. This prevents you from accidentally completing a current loop through your body if you simultaneously touch two points at different potentials.

Circuit Assembly — Step by Step

1. Connect the + **terminal** of the power supply to one lead of the 100 Ω resistor.
2. Take the **other resistor lead** back toward the − terminal, but *break* that wire mid-way and insert the **ammeter** in-line so all current passes through it.
3. Clip the **voltmeter probes** across the two leads of the resistor (do not break the main wire).

4. Set the supply to **0 V (minimum)** before calling the teacher over to check.
5. Once approved, slowly raise the voltage. Confirm both meters read positive before recording.

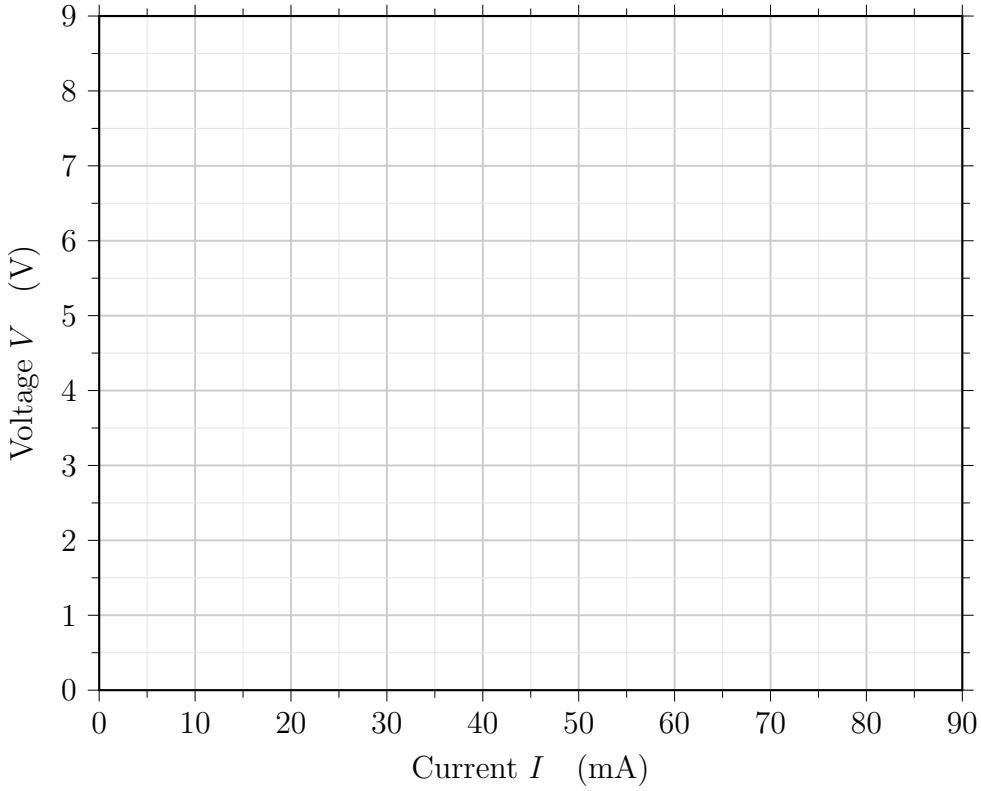
Part 1 — Data Collection Increase voltage in steps of 1.0 V. For a 100 Ω resistor, expect ≈10 mA per volt. Set the ammeter to the **mA scale**.

Voltage V (V)	**Current I (mA)**	**Notes / observations**
1.0		
2.0		
3.0		
4.0		
5.0		
6.0		
7.0		
8.0		

Part 2 — Graph: V vs. I

Plot your data: V (volts) on the vertical axis, I (milliamps) on the horizontal axis. Then, draw a **best-fit straight line** through your data points.

Expected values for $R = 100\,\Omega$: $I \approx 10\,mA$ at 1 V, rising to ≈80 mA at 8 V.



Part 3 — Analysis

1. Calculate the slope of your best-fit line including units.
   Hint: x-axis is in mA. Convert to A (divide by 1000) before computing slope.

Slope = _____

2. What physical quantity does the slope represent? Justify your answer using $V = IR$.

3. Is your graph a straight line through the origin? What does this tell you about whether the resistor is *ohmic*? What would a *non-ohmic* V–I curve look like?

4. **Prediction connecting to Section 3:**
   If you added a second $100\,\Omega$ resistor in **series**, what would R_{eq} be, and how would the slope of the V–I graph change?

   What if the second resistor were in **parallel** instead?

5 BIG IDEAS: PUTTING IT ALL TOGETHER

From mechanics to circuits — the same physics, a new arena:

- **Voltage** ΔV: electric PE per unit charge ($\Delta V = \Delta U_E/q$), just as gh is gravitational PE per unit mass. The battery pumps charges back up to maintain a constant height difference.
- **Current** I: charge flow rate ($I = \Delta q/\Delta t$). Larger ΔV at fixed $R \Rightarrow$ larger I.
- **Resistance** R: opposes current, like a narrow pipe impedes water. Larger R at fixed $\Delta V \Rightarrow$ smaller I.
- **Ohm's Law** $V = IR$: an ohmic device gives a *linear* V–I graph passing through the origin; the slope equals R.
- **Series:** $R_{\text{eq}} = R_1 + R_2$ (adding resistor increases resistance). Same I.
- **Parallel:** $\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2}$ (adding resistor decreases resistance). Same ΔV.

Looking ahead: What happens when a circuit has *multiple loops* and *multiple sources*? That's where **Kirchhoff's Laws** come in — two conservation laws you already know from mechanics applied to circuits.

Quantity	Symbol	Unit (full name)	Measured by
Current	I	ampere (A)	_____
Voltage	V, ΔV	_____	_____
Resistance	R	_____	calculated from V/I

Lab self-check:

Slope of my V–I graph = _____. Units: _____ per _____ = _____ ✓

My measured R: _____ Ω. Labelled value: _____ Ω. How close?